The Decay Envelope of the Rvachev Sinc Product: Second-Wave Audit

The unrestricted Cesàro question resolved, four syntheses verified, and corrections to the first audit — including to itself

Second comparative analysis for Vladimir Reshetnikov's `ProveIt` project
covering the four drafts `rvachev_up_fourier_decay-5` through `-8`

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Abstract

Four further documents on the Fourier decay of $f(x) = \prod_{n \geq 0} \operatorname{sinc}(\pi x/2^n)$ arrived after, and in explicit response to, the comparative audit of the first four [2]. All four are syntheses of the whole corpus, and all four are essentially correct. This document audits them, verifies their new claims, and records what they change.

The mathematical event of the wave is the resolution of the one question the first audit left open — and whose answer it guessed wrongly in one sentence. Documents 7 and 8 prove, by two independent constructions, that the *unrestricted* Cesàro limit $\frac{1}{t-t_0}\int_{t_0}^{t} |f|/g \to 1$ *is* attainable: there exists a positive, real-analytic, strictly decreasing gauge g with $\log g(x) = -(\log x)^2/(2\log 2) - \kappa_1 \log x + o(\log x)$ attaining the limit for all t. The mechanism — equalizing the mass of the shell measures on slowly refining dyadic partitions, hiding the required upward corrections under the within-shell baseline slope $-(k+\kappa_1+1)/2$, and finishing with the Hoischen–Carleman theorem on simultaneous entire approximation of a function and its derivative — survives a line-by-line adversarial audit here. Document 7 also strengthens the first audit's obstruction from shell-shape-stabilizing gauges to all bounded-distortion gauges, so the final dichotomy is: every stabilizing, bounded-distortion, or Hardy-field gauge fails (Q2); a shell-adaptive analytic gauge with necessarily unbounded, ever finer within-shell structure attains it.

Document 6 contributes a one-line martingale decomposition ($\mathcal{P}\psi = \psi/2$, $d = 2\psi - \psi \circ T = \log|\tan \pi t|$, $\int_0^1 d^2 = \pi^2/4$ exactly) that makes the corrected iterated-logarithm constant self-contained, and the exact peak-ray identity $|f(2^n \cdot \frac{2}{3})| = C_* \, E_{\kappa_\infty}(2^n \cdot \frac{2}{3})$ for all $n \geq 0$, verified here to 38 digits with $C_* = 0.13912977473482934529\ldots$ Document 5 sharpens the Gelfond bound to $\|P_n\|_\infty \leq (\sqrt{3}/2)^{n-1}$ and located the narrow excursions of the Cesàro mantissa profile that the first audit's 0.025-step grid missed.

The audit recomputes the corpus constants at forty working digits: $\varrho_1 = 0.66132260206056561507352 91\ldots$, $A_1 = 0.09126612413158821228 66\ldots$, $A_1^{\log} = 0.09386063575678126354 07\ldots$ Five small numerical errata in the wave are identified (last digits of B_1 and $A_1^{\log}$ prints, over-precise direct-quadrature digits, finite-level feature values evaluated at rounded cell boundaries), plus one shared artifact inherited by *every* document including the first audit: the sixteen-digit print $\kappa_1 = 2.748070014871334$ carries the truncation of ϱ_1 at fifteen digits; the resolved value is $\kappa_1 = 2.7480700148713326739\ldots$ The first audit's own errata are itemized: the phrase "unattainable in principle" (its theorem was correctly scoped; the slogan was not), the grid-limited profile range $[0.9249, 1.1133]$ (true features reach $\approx [0.91627, 1.11814]$), and the constant $A_1^{\log} = 0.0938610$ (seventh-figure error from a finite-shell running mean).

Contents

1 Purpose, inputs, and method

This document audits the four second-wave drafts

label	file	leng
Document 5	`rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex`	2043
Document 6	`rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex`	2474
Document 7	`rvachev_up_fourier_decay-7/rvachev_up_fourier_decay_final.tex`	1842
Document 8	`rvachev_up_fourier_decay-8/rvachev_up_fourier_decay_final.tex`	1566

All four were written with access to the first-wave documents 1–4, to the local copies of Aistleitner–Hofer–Larcher [3] and Fouvry–Mauduit [5], and to the first audit [2]; each positions itself as a final synthesis and each contains an explicit assessment of the first audit. The present document is therefore simultaneously an audit of the four drafts and a self-audit: where the second wave corrects the first audit, the correction is verified and, if it survives, accepted explicitly in section 6.

The method is that of the first audit. Every adjudication rests on a proof or an identified proof gap; numerics serve as insight and as validation. All quantitative claims of the wave that admit computation were recomputed independently, at forty working digits where precision matters (`mpmath` spectral collocation, script `stage7_spectral.py`) and with integer-exact dyadic argument reduction in the finite-level quadratures (script `stage8_profile.py`); the scripts live in `verification_scripts/` beside this file.

Throughout, $a = \log 2$, $E_\kappa(x) = x^{-\kappa}\exp(-(\log x)^2/(2a))$, $s(t) = |\sin \pi t|$, $Tt = 2t \bmod 1$, $P_n(t) = \prod_{j<n} s(T^j t)$, and $(\mathcal{L}_q h)(x) = \frac{1}{2}[\sin^q(\frac{\pi x}{2})h(\frac{x}{2}) + \cos^q(\frac{\pi x}{2})h(\frac{x+1}{2})]$, with Perron roots ϱ_q, scales $\Lambda_q = \varrho_q^{1/q}$, and exponents $\kappa_q = \frac{1}{2} + \log_2(\pi/\Lambda_q)$. The two forms of the original question [1] are, for a positive, eventually analytic, eventually decreasing gauge g,

$$\frac{1}{t-t_0}\int_{t_0}^{t} \frac{|f(x)|}{g(x)}\,\mathrm{d}x \longrightarrow 1 \tag{Q2}$$

and the two-sided bounded relaxation of the same ratio

$$0 < A \leq \frac{1}{t-t_0}\int_{t_0}^{t} \frac{|f(x)|}{g(x)}\,\mathrm{d}x \leq B < \infty. \tag{Q3}$$

Verdict at a glance

document	principal novelty	verdict
Document 5	sharp $\|P_n\|_\infty \le (\sqrt{3}/2)^{n-1}$; refined constants; located the narrow profile features; honest scoping of the open question	essentially fully correct; two last-digit numerical errata (theorems 5.1 and 5.5)
Document 6	one-line Gordin decomposition $d = \log\|\tan \pi t\|$; exact peak-ray constant C_*; extended q-spectrum table; scope-disciplined reconciliation	essentially fully correct; every checked number exact to print
Document 7	bounded-distortion obstruction; *the adaptive existence theorem* for (Q2); refined profile extremes; two-method constant computation	correct, including the central new theorem; one wrong 14th digit and over-precise direct-quadrature prints (theorems 5.2 and 5.3)
Document 8	independent second proof of the adaptive existence theorem; double-angle proof of $\mathcal{P}X = X/2$; precise mapping of the published-paper correction; Lean-friendly decomposition	essentially fully correct; constants appendix exact except the shared κ_1 artifact (theorem 5.4)

The corpus is now internally consistent: no substantive mathematical disagreement remains among documents 1–8, the two audits, and the corrected edition of [3]. The one first-wave conflict (the LIL constant) stays resolved as in the first audit — variance $\pi^2/4$ per dyadic level, constant $\pi/\sqrt{2}$ in the $\sqrt{n \log\log n}$ normalization — and all four second-wave documents state it correctly, three of them with new self-contained proofs.

2 The state of the problem after the first audit

The first audit [2] established, with proofs: the exact dyadic-shell factorization; the decay spectrum $\kappa_\infty < \kappa_2 < \kappa_1 < \kappa_0$; that $g = A_1 E_{\kappa_1}$ answers (Q3) and attains (Q2) along $t = 2^k$; that the limit points along $t = 2^k y$ trace a continuous nonconstant mantissa profile $\mathscr{P}(y) = (1 + \mathscr{F}(y-1))/y$ generated by a singular transfer eigenmeasure; that consequently no gauge with a *stabilizing* within-shell shape — in particular no Hardy-field gauge — attains (Q2); that the logarithmic mean repairs (Q2) exactly; and the corrected iterated-logarithm constant, with the published-paper erratum.

Three loose ends remained.

1. **The unrestricted question.** Does *some* positive analytic decreasing gauge attain (Q2), stabilization not assumed? The first audit's theorem excluded only the stabilizing class; its summary sentence nevertheless called (Q2) "unattainable in principle." All four second-wave documents flag the overreach; Documents 7 and 8 settle the question — affirmatively (section 3).
2. **The profile's true range.** The first audit reported the mantissa profile to span $[0.9249, 1.1133]$, measured on a step-0.025 grid. Documents 5 and 7 resolve narrow excursions the grid missed (section 5.2).
3. **Constant precision.** The corpus carried ϱ_1 to fifteen digits and the derived constants to ten–fifteen, with small discrepancies in final digits between documents. These are adjudicated at forty working digits in section 5.

3 The resolution of the unrestricted Cesàro question

3.1 The obstruction, upgraded

Write $g_1 = E_{\kappa_1}$ and, on the shell $x = 2^k(1+z)$, $\alpha_k(\mathrm{d}z) = \varrho_1^{-(k+1)} W(z) P_{k+1}(z)\,\mathrm{d}z$ for the exact normalized shell density, so that $\alpha_k \Rightarrow \alpha$ with α a finite measure of total mass A_1, atomless, fully supported, and singular (first audit, Prop. 5.3–Thm. 5.4; re-proved in all four drafts). If (Q2) holds for g, then subtracting the asymptotics at $2^k(1+z)$ and 2^k forces the *shell linearity*

$$2^{-k} \int_{2^k}^{2^k(1+z)} \frac{|f|}{g}\,\mathrm{d}x \longrightarrow z \quad \text{uniformly in } z \in [0,1]. \tag{1}$$

Document 7 extracts from this the strongest class-restricted impossibility statement in the corpus:

Theorem 3.1 (bounded-distortion obstruction; Document 7, "bounded-distortion obstruction")**.** *Suppose there are constants* $0 < m \le M < \infty$ *and scalars* $c_k > 0$ *with* $m \le g(2^k(1+z))/(c_k g_1(2^k(1+z))) \le M$ *for all large* k *and all* $z \in [0,1]$*. Then* g *cannot satisfy* (Q2)*.*

Audit. The proof is a domination argument: the normalized shell measures of $|f|/g$ have the form $q_k\,\mathrm{d}\alpha_k / \int q_k\,\mathrm{d}\alpha_k$ with $M^{-1} \le q_k \le m^{-1}$; the denominators are bounded below by $M^{-1}(A_1 + o(1))$, so every weak limit is dominated by a constant multiple of the singular measure α and is therefore singular — while (1) forces the limit to be Lebesgue measure. Correct. This strictly subsumes the first audit's stabilizing-shape theorem (a uniformly convergent positive shape is eventually a bounded distortion), and it disposes of gauges the stabilizing theorem never covered. Document 7's example

$$g_{\varepsilon,c}(x) = g_1(x)\exp\bigl(\varepsilon \sin(c(\log x)^2)\bigr), \qquad 2c\varepsilon < 1/\log 2,$$

is elementary, positive, analytic, eventually strictly decreasing (the stated parameter condition is exactly what eventual monotonicity requires; verified), and *not* of Hardy-field type — its shell shape never stabilizes — yet it still fails (Q2) by theorem 3.1. This example also corrects the first audit's parenthetical equation of "Hardy field" with "elementary" (item 2 of section 6).

3.2 The existence theorem

Theorem 3.2 (adaptive analytic gauge; Document 7, "analytic exact Cesàro gauge", and Document 8, "literal analytic solution of the original limit problem", independently)**.** *There exists a positive, real-analytic, strictly decreasing function* g *on a half-line (Document 8's variant: on all of* $(0,\infty)$*) such that* (Q2) *holds for every fixed* t_0*, and*

$$\log g(x) = -\frac{(\log x)^2}{2\log 2} - \kappa_1 \log x + o(\log x).$$

By theorem 3.1, any such gauge necessarily has unbounded *within-shell distortion relative to* g_1*: the correction it carries cannot stabilize, cannot be bounded, and cannot be a fixed elementary factor.*

The two proofs share one mechanism and differ in bookkeeping. Both were audited line by line; both are correct. Because this theorem reverses the first audit's slogan, the audit trail is recorded in detail.

The mechanism

The within-shell slope of the baseline is enormous. In shell coordinates $x = 2^k(1+z)$,

$$\frac{\mathrm{d}}{\mathrm{d}z} \log g_1\bigl(2^k(1+z)\bigr) = -\frac{k + \kappa_1 + \log_2(1+z)}{1+z} \;\le\; -\frac{k+\kappa_1+1}{2} \;=:\; -2b_k, \tag{2}$$

the extreme value being attained at $z = 1$ (verified by direct differentiation of $\log g_1 = -L^2/(2a) - \kappa_1 L$, $L = ka + \log(1 + z)$). Any multiplicative correction whose logarithmic slope is $o(k)$ — in Document 7's version, bounded below by $-b_k$ — therefore rides on the baseline without disturbing strict monotonicity. The obstruction of section 3.1 only forbids *bounded* corrections; a correction of sup-norm $o(k)$ in the logarithm is unbounded over the family of shells while still asymptotically invisible at the $\log g$ scale, and that is exactly the room the construction uses.

1. **Step equalizer.** Fix a dyadic partition depth r, $\delta = 2^{-r}$. On each cell $I_{r,j}$ set $q \equiv \delta/\alpha_k(I_{r,j})$. Then every cell of the reweighted measure $q\, \mathrm{d}\alpha_k$ carries mass exactly δ, so its cumulative function is within δ of z. For fixed r the cell values and their reciprocals stay bounded as $k \to \infty$, because $\alpha_k(I_{r,j}) \to \alpha(I_{r,j}) > 0$ (weak convergence against an atomless, fully supported limit; cell boundaries are α-null).

2. **Smoothing under the slope budget.** The step jumps are replaced by C^∞ bridges supported in shrinking neighborhoods of the finitely many cell boundaries, with $q \equiv 1$ near both shell endpoints so that consecutive shells join as the constant 1 (g then equals g_1 near every power of two, making the global function smooth). In Document 7, a downward log-jump of size D is spread over width $> D/b_k$, keeping $(\log q_k)' \geq -b_k$, so $\frac{\mathrm{d}}{\mathrm{d}z} \log(g_1/q_k) \leq -2b_k + b_k = -b_k < 0$; upward jumps of q only steepen the decrease. In Document 8, which corrects the reciprocal $h = 1/g$, the requirement is simply $\|(\log r_k)'\|_\infty = o(k)$ against a reciprocal baseline slope $\geq k/3$. Bridge widths tend to 0 (since $b_k \to \infty$), and the α_k-mass of the bridge neighborhoods tends to 0 by atomlessness of α, so the equalization survives smoothing up to $o(1)$.

3. **Diagonalization.** Choose thresholds $K_r \uparrow \infty$ so that for $K_r \leq k < K_{r+1}$ the depth-r equalizer achieves cumulative error $\leq \delta + 1/r$ *and* its logarithmic size C_r satisfies $C_r/K_r \leq 1/r$. As $k \to \infty$ the depth $r(k) \to \infty$ slowly; the cumulative error tends to 0, and $\sup |\log q_k| = o(k)$, which is $o(\log x)$ — giving the stated rough asymptotic. (The unbounded relative variation of the cell multipliers as $r \to \infty$ is precisely the singularity of α expressing itself; the slow refinement keeps it below the slope budget.)

4. **Summation.** With per-shell uniform equalization error $\varepsilon_k \to 0$, partial-shell integrals are $2^k(z + \mathcal{O}(\varepsilon_k))$ and the elementary Toeplitz bound $\sum_{j<K} 2^j \varepsilon_j = o(2^K)$ gives $\int_{t_0}^t |f|/g_0 = t + o(t)$ for the smooth gauge g_0 — i.e., (Q2) with limit 1, for *all* t.

5. **Analytic upgrade.** Both documents invoke the classical Carleman-type theorem of Hoischen (in the modern form of Gauthier–Kienzle [7, 6]): for $F \in C^1(\mathbb{R})$ and any continuous $\epsilon > 0$ there is an entire G, real on $\mathbb{R}$, with $|G - F| < \epsilon$ and $|G' - F'| < \epsilon$ everywhere. Document 7 applies it to $F = \log g_0$ with $\epsilon < \min(-F'/2,\ \text{something} \to 0)$, obtaining $g = \mathrm{e}^G$ zero-free, positive, strictly decreasing, with $g/g_0 \to 1$; Document 8 applies it to the increasing reciprocal h with $\epsilon < \frac{1}{2}\min(h, h', \delta h)$, obtaining $\widetilde{h} > 0$, $\widetilde{h}' > 0$ on all of $\mathbb{R}$ and $\widetilde{h}/h \to 1$, hence a gauge positive and strictly decreasing on all of $(0, \infty)$. In both cases $g/g_0 \to 1$ preserves the Cesàro limit by a split-and-squeeze argument (checked).

Points probed and found sound

The audit specifically attacked, and failed to break, the following potential gaps. (i) *Monotone matching of an oscillating target*: a decreasing gauge can only lower its level, so cell corrections that must move up and down seem incompatible with monotonicity — this is where (2) is decisive, and the numerical margin ($2b_k$ available, b_k used) is explicit. (ii) *Convergence of cell masses*: needs $\alpha(\partial I) = 0$, supplied by atomlessness. (iii) *Positivity of finite-level cell masses*: the density $W\, P_{k+1}$ vanishes only on a finite set, so $\alpha_k(I) > 0$ for every cell at every finite k. (iv) *Bridge mass*: finitely many shrinking intervals, uniformly bounded q, and an atomless limit give $o(1)$; the interior partition boundaries are even zeros of P_{k+1}, making the perturbation smaller still (noted by Document 7). (v) *Shell joins*: $q \equiv 1$

zones make $g = g_1$ in two-sided neighborhoods of the powers of two; g_1 is decreasing there. (vi) *The Hoischen error function*: $-F'/2$ is continuous and positive on the half-line, so an admissible ϵ exists; the approximation preserves both signs and the limit. No gap was found.

Remark 3.3 (why this does not contradict the singular profile). At every *fixed* partition depth the flattened shell measures converge weakly to Lebesgue; but the reweighting has no bounded limiting density with respect to the singular α — its relative variation blows up as the depth refines. The obstruction theorems exclude precisely the gauges that cannot do this. Both documents make this point; Document 8's formulation ("the number of corrected cells tends to infinity, the correction's log-slope is $o(k)$") is the cleanest.

3.3 The final map of the original question

The literal answer to [1], final form. With $g_1 = E_{\kappa_1}$, $\kappa_1 = \frac{1}{2} + \log_2(\pi/\varrho_1)$:

- (Q3): **yes**, with $g = A_1 g_1$, $A_1 = 0.09126612413158821\ldots$; the limit points are exactly the range of the continuous singular profile $\mathscr{P}$, numerically $\approx [0.91627, 1.11814]$ (section 5.2).
- (Q2) along $t = 2^k$: **yes**, same gauge, limit 1.
- (Q2) unrestricted, within any stabilizing, bounded-distortion, or Hardy-field class: **no** (theorem 3.1).
- (Q2) unrestricted, over all positive analytic decreasing gauges: **yes** (theorem 3.2) — by a shell-adaptive construction whose within-shell microstructure necessarily refines forever; no fixed formula does it.
- Multiplicative repair: with dx/x averaging, $A_1^{\log} g_1$ attains the exact unrestricted limit, $A_1^{\log} = 0.09386063575678126\ldots$ (first audit's theorem; refined constant).

This composite answer supersedes item (1) of the first audit's verdict: where that verdict said "unattainable for any gauge in the natural class," the class restriction is now known to be essential, not cosmetic.

4 Document-by-document audit

4.1 Document 5

A synthesis in the style of the first audit, with a claims-first layout. Everything checked agrees with the corpus: the spectrum, the shell factorization, the singular-profile obstruction (stated for the stabilizing class, with an explicit and correct warning that the unrestricted class "is not eliminated by the supplied arguments and should not be claimed to be eliminated"), the logarithmic-mean repair, and the corrected LIL with the right attribution to the published paper's Parseval slip.

Novelties verified.

1. **Sharp uniform bound.** Document 5 (and, with the same two-step inequality, Document 6) improves the calibrated Gelfond bound $P_n \le \sqrt{5/3}\,(\sqrt{3}/2)^n$ of the first wave to
$$\|P_n\|_\infty \le \left(\tfrac{\sqrt{3}}{2}\right)^{n-1},$$
from the weighted two-step inequality $\frac{2}{3}\log s(x) + \frac{1}{3}\log s(Tx) \le \log\frac{\sqrt{3}}{2}$, with equality iff $x \equiv \frac{1}{3}, \frac{2}{3} \pmod 1$. Audit: cubing reduces the inequality to $s(x)^2 s(Tx) \le 3\sqrt{3}/8$, i.e. to $\max_{u\in[0,1]} u^3(1-u) = 27/256$ at $u = \sin^2 \pi x = 3/4$; the telescoping bookkeeping $\sum \phi_j \le (n-1)\log\lambda + \frac{1}{3}\phi_0 + \frac{2}{3}\phi_{n-1}$ uses $\phi \le 0$. Correct, elegant, and sharp at both ends ($\|P_1\|_\infty = 1 = \lambda^0$; the value λ^n is attained at $t = \frac{1}{3}$). A natural Lean target (section 7).

2. **Refined constants.** $A_1 = 0.0912661241315\ldots$ (correct; section 5.1), and the logarithmic-shell pair B_1, $A_1^{\log} = B_1/a$ — correct to thirteen digits, with a last-two-digit erratum in the B_1 print (theorem 5.1).

3. **Profile features.** The observation that the first audit's $[0.9249, 1.1133]$ was a grid artifact, and the location of narrow excursions at mantissa $y = 8/7$ and $y = 10/7$. Qualitatively confirmed and quantitatively refined here (section 5.2); the printed finite-level values carry a small evaluation-scheme offset (theorem 5.5).

4. **Lambert finite-shell check.** Document 5's residual table for the annular-maximum formula matches the first audit's own table after the centering offset $\beta - \log \pi$ is accounted for (checked during the first reading; e.g., at $k = 8$ their -2.00423 against the audit's -0.7157 shifted by -1.2886).

Criticisms of the first audit. Two are raised: the over-broad "in principle" phrasing, and the under-resolved profile range. Both are fair and both are accepted (section 6).

Verdict. Essentially fully correct; the most careful of the four about what remained open — it explicitly declined to guess the answer that Documents 7–8 later proved.

4.2 Document 6

The Document-4 lineage synthesis: transfer-operator first, with the Collatz–Wielandt/Sturm certificate for ϱ_1 reproduced verbatim from Document 4 (the certificate the first audit already verified by exact Sturm sequences; interval $(0.66126807, 0.66134891)$), alongside the published sharper enclosure $0.66130 < \varrho_1 < 0.66135$ correctly attributed to [3].

Novelties verified.

1. **One-line Gordin decomposition.** With $\psi(t) = \log|2 \sin \pi t|$ and $\mathcal{P}$ the unweighted Perron operator of doubling, the Clausen expansion $\psi = -\sum_m \cos(2\pi m t)/m$ gives $\mathcal{P}\psi = \psi/2$; hence

$$d := 2\psi - \psi \circ T = \log|\tan \pi t|, \qquad \mathcal{P}d = 0, \qquad S_n = \sum_{j<n} d \circ T^j - \psi + \psi \circ T^n,$$

and $\int_0^1 d^2 = \frac{2}{\pi} \int_0^{\pi/2} \log^2(\tan u)\, \mathrm{d}u = \pi^2/4$ *exactly* — the martingale variance equals the asymptotic variance with no covariance corrections. Audit: the identity $2\psi - \psi \circ T = \log|\tan \pi t|$ is the double-angle formula; $\mathcal{P}d = 0$ is $\tan\theta \cdot \tan(\theta + \frac{\pi}{2}) = -1$; the integral is the classical $\int_0^{\pi/2} \log^2 \tan = \pi^3/8$, which the verification suite reproduces to twenty digits. The reverse-martingale CLT/LIL then imports only Stout and Heyde–Scott, and the coboundary terms are handled by exponential tails and Borel–Cantelli. Correct, and the shortest self-contained route to the corrected constant in the whole corpus.

2. **Exact peak ray.** For every $n \geq 0$, $|f(2^n \cdot \frac{2}{3})| = C_* E_{\kappa_\infty}(2^n \cdot \frac{2}{3})$ exactly, with $C_* = W_{\kappa_\infty}(2/3) = 0.139129774734829\ldots$ Audit: on the ray the doubling orbit of the mantissa is exactly the period-two cycle $\{\frac{1}{3}, \frac{2}{3}\}$, every sine factor is exactly $\frac{\sqrt{3}}{2}$, and the shell factorization turns the statement into an identity — there is nothing asymptotic about it. Verified numerically at $n = 0, 5, 10, 15$: the ratio is constant to $2.8 \cdot 10^{-38}$, value $C_* = 0.13912977473482934529458\ldots$; all fifteen printed digits correct (section 5.3).

3. **Extended q-spectrum.** The collocation values $\varrho_3, \varrho_4, \varrho_8, \varrho_{16}$ and their κ_q; all reproduced to every printed digit (section 5.1; the ϱ_3, ϱ_4 rows also match the first audit's own table).

4. **Finite-level profile table.** Sixteen values of $\mathscr{P}$ at level 18; all sixteen reproduced here to $4 \cdot 10^{-10}$ (section 5.2).

5. **Exact variance bookkeeping.** The finite-n formula $\int S_n^2 = \frac{\pi^2}{4}n - \frac{\pi^2}{3}(1-2^{-n})$ re-derived via $C_r = \pi^2/(12\cdot 2^r)$ and $\sum_{r=1}^{n-1}(n-r)2^{-r} = n-2+2^{1-n}$ (identity checked); agrees with the first audit's theorem.

Scope discipline. Document 6's reconciliation section states precisely that the obstruction covers the stabilizing class and that "none of the source materials proves such a fully unrestricted nonexistence theorem" — correct at its date of writing, superseded by Documents 7–8 within the same wave.

Verdict. Essentially fully correct; every checked number exact to print; the fluctuation section is the corpus's cleanest.

4.3 Document 7

The most consequential document of the wave: it contains the bounded-distortion obstruction (theorem 3.1), the adaptive existence theorem (theorem 3.2), and the sharpest finite-level profile numerics.

Additional items verified.

1. **RPF architecture.** Its Lasota–Yorke constants for $\mathcal{L}_1$ on Lipschitz functions ($\|\mathcal{L}\varphi\|_\infty \le 2^{-1/2}\|\varphi\|_\infty$; contraction factor $\sqrt{2}/4$ for the Lipschitz seminorm) are correct, and its lower bound $\varrho_1 \ge 1/\sqrt{3} > \sqrt{2}/4$ via Cauchy–Schwarz, the exact $\int P_n^2 = 2^{-n}$, and the calibrated Gelfond bound is exactly the first audit's bracket (the part already formalized in Lean, section 7). The essential-radius argument makes the leading spectrum isolated without importing Fouvry–Mauduit; the simplicity/irreducibility step is compressed but standard. Among all eight documents this comes closest to a self-contained RPF package.

2. **Hardy-field paragraph.** Matches the first audit's dichotomy ($xh'(x) \to \gamma$ finite $\Rightarrow$ stabilizing shape; infinite $\Rightarrow$ endpoint concentration), now embedded in the stronger bounded-distortion framework.

3. **Lambert and valley sections.** The annular-maximum theorem (with the same $c = (a/\pi)\sqrt{3/2}$ and $-(w^2+2w)/(2a)$ correction) and the two-adic peak law $E_{m2^k} \sim H(1)|H(m)|(m2^k)^{-(k+1)}/(\mathrm{e}(k+1))$ agree with Documents 2/4 and with the first audit's own §6 (the expansion coefficients in its "expanded" form were re-derived here from the W-asymptotics and check out, including the $-(1+\log c)\log k/a$ term).

4. **Two-method constants.** Spectral collocation and direct cell quadrature agreeing to ~ 13 digits on A_1; adjudicated at higher precision in section 5.1 (the spectral value is right; the direct print's tail digits are quadrature artifacts, theorem 5.3).

5. **Its critique of the first audit** (three items: profile range, "Hardy-field hence elementary," "unattainable in principle"). All three are fair; all three are accepted in section 6.

Verdict. Correct, including the central new theorem; the two numerical blemishes (theorems 5.2 and 5.3) do not touch any mathematical statement.

4.4 Document 8

An independent synthesis converging on the same final picture as Document 7, including its own proof of the adaptive existence theorem — found independently, with a different and arguably cleaner technical route (flattening multipliers r_k on the *reciprocal* $h = 1/g$, slope budget $\|(\log r_k)'\|_\infty = o(k)$ against a reciprocal baseline slope $\ge k/3$, Hoischen applied to h with $\epsilon < \frac{1}{2}\min(h, h', \delta h)$, which yields strict decrease on all of $(0,\infty)$). The flattening lemma is stated as a standalone measure-theoretic fact about weakly convergent sequences of measures with atomless fully supported limits — the right level of generality, and the version most suitable for formalization.

Additional items verified.

1. **Double-angle Perron identity.** $\mathcal{P}X = X/2$ for $X = \log|2\sin\pi t|$ proved from $2\sin\frac{\pi x}{2} \cdot 2\cos\frac{\pi x}{2} = 2\sin\pi x$ — no Fourier series needed; and $\int d^2 = 5c_0 - 4c_1 = \pi^2/4$ by pure covariance algebra ($c_0 = \pi^2/12$, $c_1 = \pi^2/24$). Both check.
2. **Precise erratum mapping for [3].** Its list — (4.5) $\sigma^2 = \pi^2/2 \to \pi^2/4$; Lemma 4.1 constant $\pi/\sqrt{2} \to \pi/2$; Theorem 1.2 constant $\pi \to \pi/\sqrt{2}$; Theorem 1.1 constant $\pi/\sqrt{\log 2} \to \pi/\sqrt{2\log 2}$; Fouvry–Mauduit constant and L^1 discrepancy exponent unaffected — coincides exactly with the first audit's findings and with the corrected edition [4].
3. **Master normalization.** Its identity $|f(2^k y)|/E_\kappa(2^k y) = C_\kappa(y)\exp(k(\kappa a - \log\pi - a/2))P_{k+1}(y-1)$ was re-derived here from scratch; the bookkeeping (including the $\kappa = \kappa_0$ specialization with $D_0 = C_{\kappa_0}/2$) is exact.
4. **Constants appendix.** All nineteen–twenty digit values reproduced exactly — except the κ_1 entry, which carries the corpus-wide truncation artifact (theorem 5.4).
5. **Lean-friendly decomposition.** Its eight-block plan for formalization is sensible; blocks 1 and 3 are already done in the repository (section 7).

Verdict. Essentially fully correct; together with Document 7 it settles the unrestricted question beyond reasonable doubt — two independent proofs, one mechanism.

5 Numerical adjudication

Two independent pipelines were used. *Spectral* (`stage7_spectral.py`): Chebyshev–Lobatto collocation of $\mathcal{L}_q$ at $N = 48$ and $N = 64$ nodes in 40-digit arithmetic; leading triple by power iteration on M and M^T; $\int h$ by Clenshaw–Curtis on the same nodes; $A_1 = (\int h)\,\nu(W)$, $B_1 = (\int h)\,\nu(W/(1+\cdot))$ with the discrete left eigenvector as the eigenmeasure's atom weights. The two node counts agree to 10^{-38}. *Direct* (`stage8_profile.py`): 16-point Gauss–Legendre quadrature on each of the 2^n dyadic cells of the shell, with the dyadic part of every $2^j z \bmod 1$ reduced in exact int64 arithmetic — which is what lets the direct values below agree with the spectral anchor to ≈ 16 digits instead of the ≈ 13 visible in the wave's own float64 pipelines.

5.1 The constants

quantity	value (this audit)
ϱ_1	$0.66132260206056561507352912 15\ldots$
κ_1	$2.7480700148713326739171436\ldots$
A_1	$0.0912661241315882122865631\ldots$
B_1	$0.0650592350403769214309905\ldots$
$A_1^{\log} = B_1/a$	$0.0938606357567812635406834\ldots$
C_*	$0.1391297747348293452945836\ldots$
$\varrho_3,\ \kappa_3$	$0.39489663237023\ldots,\quad 2.59831380619\ldots$
$\varrho_4,\ \kappa_4$	$0.3201941016011\ldots,\quad 2.56224147026\ldots$
$\varrho_8,\ \kappa_8$	$0.16124445889978\ldots,\quad 2.48058094336\ldots$
$\varrho_{16},\ \kappa_{16}$	$0.05007193571264\ldots,\quad 2.42148700201\ldots$

The direct pipeline confirms the spectral A_1 and $A_1^{\log}$:

level	$A_1^{(n)}$	$A_1^{\log\,(n)}$
$n = 13$	0.09126612412933643	0.09386063575634529
$n = 15$	0.09126612413169176	0.09386063575669520
$n = 17$	0.09126612413160065	0.09386063575677621
$n = 18$	0.09126612413158525	0.09386063575678234
$n = 19$	0.09126612413158888	0.09386063575678112
$n = 20$	0.09126612413158816	0.09386063575678139
spectral	0.09126612413158822	0.09386063575678126

Against these anchors, the wave's prints adjudicate as follows. A_1: Document 5's thirteen digits, Document 6's fifteen digits (0.091266124131588), and Document 7's rounded fourteen (0.09126612413159) are all correct. C_*, the q-spectrum rows, Document 6's ϱ_1-collocation convergence table (reproduced here digit for digit at every N), Document 7's variance check $\mathrm{Var}(S_{1000}) = 2464.1112321386$, and all of Document 8's appendix values except κ_1 are correct. The errata:

Erratum 5.1 (Document 5's B_1 print). Document 5 prints $B_1 = 0.065059235040389\ldots$ and, in its numerics section, $B_1/a = 0.093860635756799\ldots$ The resolved values are $B_1 = 0.06505923504037692\ldots$ and $B_1/a = 0.09386063575678126\ldots$: the fourteenth–fifteenth digits of both prints are wrong (its separate thirteen-digit print $A_1^{\log} = 0.0938606357568$ is correct).

Erratum 5.2 (Document 7's spectral $A_1^{\log}$). Document 7 prints $A_1^{\log} = 0.09386063575679\ldots$; the fourteenth digit should be 8, not 9 (its own direct-quadrature print 0.09386063575678312 has it right).

Erratum 5.3 (over-precise direct-quadrature digits). Document 7's direct value $A_1^{(18)} = 0.09126612413156757$ (17 digits) and Document 5's analogous convergence table carry $\sim 2 \cdot 10^{-14}$ of float64 argument-reduction noise in their final digits: the exact level-18 value is $0.091266124131585\ldots$ Nothing downstream is affected; the prints simply exceed the pipelines' precision.

Erratum 5.4 (the shared κ_1 print). Every document of both waves — *including the first audit's abstract* — prints $\kappa_1 = 2.748070014871334$ (Document 8: $\ldots 340157$ to twenty digits; the first audit's constants table, alone in the corpus, bracketed the uncertainty as $2.748070014871334(4)$, an interval that does contain the true value). All of these were computed from ϱ_1 truncated at fifteen digits. With the resolved $\varrho_1 = 0.66132260206056561507\ldots$ the correct value is

$$\kappa_1 = 2.7480700148713326739\ldots,$$

i.e. the shared sixteen-digit print is wrong in its last two digits (correctly rounded: 2.748070014871333). The sensitivity is $\mathrm{d}\kappa_1/\,\mathrm{d}\varrho_1 = -1/(a\varrho_1) \approx -2.18$, so the $6 \cdot 10^{-16}$ truncation of ϱ_1 moved κ_1 by $1.3 \cdot 10^{-15}$ — harmless everywhere, but worth recording since the sixteen-digit value has now been copied through eight documents.

5.2 The profile at finite level

The natural-gauge Cesàro profile $\mathscr{P}_n(y) = (1 + F_n(y-1))/y$ was recomputed at levels $n = 13, 17, 18, 19$. Three independent cross-checks tie the pipelines of Documents 6, 7, and this audit together:

- all sixteen entries of Document 6's level-18 table are reproduced to at most $4.3 \cdot 10^{-10}$;
- Document 7's level-19 boundary extremes are reproduced to all ten printed digits: minimum 0.9162686675 at $y = 1.142841339\ldots$, maximum 1.1181395810 at $y = 1.428632736\ldots$;
- Document 7's level-18 root-refined extremes (0.9162685832, 1.1181397445) lie just beyond this audit's boundary-only scan (0.9162689057, 1.1181388297), exactly as interior refinement should.

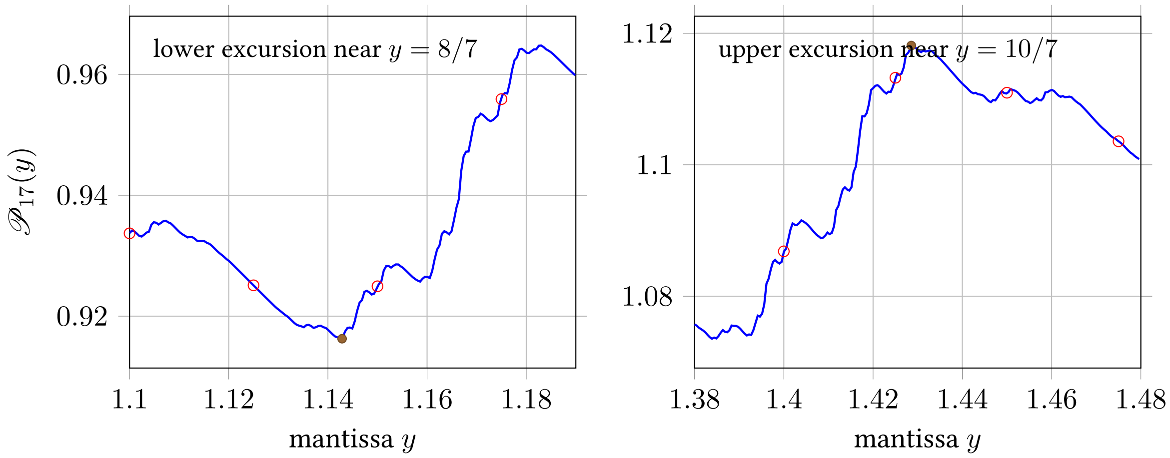

Figure 1: The two narrow excursions of the Cesàro mantissa profile at level 17 (solid; boundary values at spacing 2^{-11}), the first audit's step-0.025 grid points (circles), and the exact values at $y = 8/7$ and $10/7$ (filled dots). The 0.025 grid interpolates straight across both features — the source of the first audit's under-estimated range $[0.9249, 1.1133]$ — and even the 2^{-11} sampling does not reach the excursions' tips: the exact rational points dip visibly beyond the sampled curve, as expected of a singular-continuous limit profile.

Erratum 5.5 (Document 5's finite-level feature values). Document 5 prints $\mathscr{P}_{13}(8/7) = 0.9163240341$ and $\mathscr{P}_{13}(10/7) = 1.1181043233$ (and 0.9162741, 1.1181341 at $n = 17$). Exact partial-cell evaluation gives $\mathscr{P}_{13}(8/7) = 0.9162705095$, $\mathscr{P}_{13}(10/7) = 1.1181038033$, $\mathscr{P}_{17}(8/7) = 0.9162696223$, $\mathscr{P}_{17}(10/7) = 1.1181382158$. The offsets ($5 \cdot 10^{-5}$ at $n = 13$, $4 \cdot 10^{-6}$ at $n = 17$, shrinking with level) are consistent with evaluating the cumulative sum at the dyadic cell boundary nearest to $1/7$ resp. $3/7$ rather than at the exact point. The substance of the claim — narrow excursions near $8/7$ and $10/7$ far outside the first audit's grid range — is confirmed.

The corrected picture of the profile's extent: the level-n boundary extremes drift monotonically through $[0.91626891, 1.11813883]$ ($n = 18$) and $[0.91626867, 1.11813958]$ ($n = 19$); the limiting profile is singular-continuous, so ever finer features are expected and no certified global extrema exist. The honest statement, replacing the first audit's $[0.9249, 1.1133]$, is that the range *contains* $[0.91627, 1.11814]$ and that its endpoints are not yet certified. Figure 1 shows the two excursions against the first audit's grid.

5.3 The exact peak ray

Document 6's identity $|f(2^n \cdot \frac{2}{3})| = C_* E_{\kappa_\infty}(2^n \cdot \frac{2}{3})$, $n \geq 0$, was tested by direct 40-digit evaluation of the infinite product at $n = 0, 5, 10, 15$: the four ratios agree with one another to $2.8 \cdot 10^{-38}$, confirming exactness (not merely asymptotics), and

$$C_* = 0.139129774734829345294583 6\ldots,$$

whose first fifteen digits are Document 6's print. The identity deserves emphasis because it is the cleanest possible witness that the κ_∞ envelope is attained: a single explicit geometric ray on which the normalized transform is *constant*. It is also, with the machinery already formalized, a near-term Lean target (section 7).

6 Corrections to the first audit

For the record, the first audit's errors and imprecisions, as exposed by the wave and verified here. The companion patch updates the first audit's text in place; this section is the changelog.

1. **"Unattainable in principle."** The first audit's one-sentence summary called the unrestricted (Q2) "unattainable in principle." Its *theorem* was correctly scoped (stabilizing shapes only), and its verdict section stated the class restriction — but the slogan asserted more than was proved, and more than is true: theorem 3.2 refutes it. The corrected slogan: unattainable for every gauge whose within-shell shape stabilizes or stays within bounded distortion of the natural gauge — attainable, necessarily by unbounded shell-adaptive microstructure, in general.

2. **"Hardy-field ('elementary')".** The parenthetical implied that Hardy-field covers all elementary formulas; Document 7's oscillatory example $g_1 \exp(\varepsilon \sin(c \log^2 x))$ is elementary and non-Hardy. (It, too, fails (Q2) — by bounded distortion — so the first audit's conclusion survives for it; only the implication arrow was wrong.)

3. **Profile range.** $[0.9249, 1.1133]$ was presented with a grid caveat but used as "the" range; the true range contains $[0.91627, 1.11814]$ (section 5.2).

4. $A_1^{\log} = 0.0938610$**.** A seventh-significant-figure error: the printed value was a finite-K running mean of the shell sequence (β_k), whose $\mathcal{O}(1/K)$ Cesàro transient had not died out. Correct value $A_1^{\log} = 0.09386063575678126\ldots$; the first audit's own per-mantissa table (0.093861 at $y = 1$) was already consistent with it at its printed precision.

5. κ_1 **last digits.** Shared with the whole corpus (theorem 5.4).

Nothing else in the first audit is contradicted by the wave: its variance/LIL forensics, the singularity proof via $\varrho_1 \geq 3^{-1/2}$, the logarithmic-mean theorem, the lobe-maxima and RMS sections, and its per-document adjudications of Documents 1–4 all stand, and are independently re-derived in at least one second-wave document each.

7 Consequences for the corpus and the formalization ledger

The repository's Lean formalization campaign (`Analysis/FabiusFunction/Lean/FabiusFunction/`) already covers, in greater generality than the texts: the master lacunary orthogonality theorem behind $\int_0^1 P_n^2 = 2^{-n}$ (`LacunaryRieszIntegral.lean`: `integral_cos_mul_rieszProduct` and its corollary `integral_prod_sin_sq_two_pow`); the calibrated Gelfond bound along *every* logistic orbit and the resulting L^1 lower bracket $\varrho_1 \geq 3^{-1/2}$ (`GelfondLogisticBound.lean`: `logistic_prod_le`, `le_mul_integral_prod_abs_sin_two_pow`); and the dyadic shell factorization with its exponent bookkeeping (`GeometricScaleProducts.lean`, `SincProductShells.lean`: `norm_rvachevFourierProduct_two_pow_mul`, `shell_exponent_identity`). These are blocks 1 and 3 of Document 8's proposed decomposition, plus the harder half of its block 5.

The wave adds four attractive targets, in rough order of readiness:

1. **Sharp uniform bound** $\|P_n\|_\infty \leq (\sqrt{3}/2)^{n-1}$ via the weighted two-step inequality (Documents 5/6) — pure trigonometric algebra plus a telescoping sum; same difficulty class as the calibrated bound already formalized.

2. **The exact peak ray** $|f(2^n \cdot \frac{2}{3})| = C_* E_{\kappa_\infty}(2^n \cdot \frac{2}{3})$ — follows from `norm_rvachevFourierProduct_two_pow_mul` plus the exact period-two orbit values $|\sin(2^j \cdot \frac{2\pi}{3})| = \frac{\sqrt{3}}{2}$; a satisfying exact statement with no analysis beyond the existing modules.

3. **The martingale identity** $d = 2\psi - \psi \circ T = \log|\tan \pi \cdot|$ with $\mathcal{P}d = 0$ (Document 6) — pointwise double-angle identities; the gateway to a formal variance computation.

4. **The flattening lemma** (Document 8's standalone form) — a measure-theoretic statement about weakly convergent measures, independent of the sinc product; medium-term.

The exact variance $\frac{\pi^2}{4}n - \frac{\pi^2}{3}(1 - 2^{-n})$ still requires the Clausen expansion in L^2 and remains the hard fluctuation target; the adaptive gauge's Hoischen step would enter as a named classical import.

Postscript: ledger update (late August 2026 campaign)

All four targets above are now formalized, and the campaign went substantially further. The development (`Analysis/FabiusFunction/Lean/FabiusFunction/`, seventy-one sorry-free modules at this writing) now contains, kernel-checked:

- the sharp uniform bound $\|P_n\|_\infty \le (\sqrt{3}/2)^{n-1}$ with its full equality locus (`SharpGelfondBound`, `SharpGelfondEquality`);
- the exact peak ray in product and κ_∞-envelope form (`SincProductPeakRay`, `PeakRayEnvelope`);
- the martingale identity $\mathcal{P}d = 0$ and the halving $\mathcal{P}\psi = \psi/2$ (`DoublingCocycleIdentities`), the full Lasota–Yorke pair of the transfer operator with constants $(\alpha, \beta, C) = (\sqrt{2}/4, \sqrt{2}/2, \pi/2)$ (`TransferOperatorStep`) and its abstract Doeblin–Fortet iteration and invariant ball (`LasotaYorkeIteration`);
- **both Clausen values**: $\int_0^1 \log^2(2\sin \pi t)\,dt = \pi^2/12$ and $\sigma^2 = \int_0^1 \log^2|\tan \pi t|\,dt = \pi^2/4$, obtained through the Dirichlet-kernel cotangent pairing (`DirichletKernelCotangent`), the improper-integration-by-parts evaluation of the log-sine Fourier coefficients $-1/(2m)$ (`LogSineFourierCoefficient`), and interval Parseval with Basel (`CocycleParseval`, `LogCosineFourier`, `GordinParseval`);
- the complete covariance ladder $c_r = (\pi^2/12)\,2^{-r}$ for *every* lag over the genuine mod-one doubling map (`DoublingMapTransfer`, `CovarianceLadder`), and the martingale orthogonality of d (`MartingaleOrthogonality`);
- **the exact variance theorem itself**, stated on the normalized dyadic sine product:

$$\int_0^1 \log^2\Big|\prod_{k<n} 2\sin(\pi 2^k t)\Big|\,dt = \frac{\pi^2}{4}\,n - \frac{\pi^2}{3}\big(1 - 2^{-n}\big)$$

  (`DistanceCounting`, `BirkhoffVariance`, `BirkhoffProductBridge`) — the finite-n content of the fluctuation theorem, fully numeric;
- the flattening lemma's finite level (exact node equalization and the partial-cell mesh bound, `MeasureEqualization`) and its diagonal selection mechanism (`DiagonalSelection`);
- existence of the Perron root $\varrho_1 = \lim_n \|\mathcal{L}^n\mathbf{1}\|_\infty^{1/n} \in [1/2, \sqrt{2}/2]$ by Fekete's lemma (`TransferPositivity`, `PerronRootExistence`), and its uniqueness as an eigenvalue of positive continuous eigenfunctions by the m/M-sandwich (`PerronEigenvalueUnique`), together with the exact centering $\int_0^1 S_n = 0$ and the normalized limit $(1/n)\int_0^1 S_n^2 \to \pi^2/4$ (`BirkhoffMoments`);
- **the one-peak theorem in full** (the drafts' `thm:one-peak`): the log-factorization $\log\|\Phi(x)\| = \sum_{(h,r)} \log\big(1 - x^2/(2^h(r+1))^2\big)$ at every non-lattice real x, with the finitely many negative factors absorbed by the $\log|\cdot|$ convention (`LobeLogFactorization`); strict concavity of $\log|\Phi|$ on the central lobe $(-1,1)$ by twice-termwise differentiation of the lattice log-series (`CentralLobeConcavity`, `CentralLobeOnePeak`); the peak pinned at the origin with $\Phi(0) = 1$ and $|\Phi(x)| < 1$ on $(-1,1) \setminus \{0\}$, by evenness and the strict midpoint inequality alone (`CentralLobePeakAtZero`); and strict concavity of $\log\|\Phi\|$ on *every* side lobe $\pm(m, m+1)$, via a single uniform quadratic gap $\min\big((u^2 - m^2)/(m^2+1),\ 1 - v^2/(m+1)^2\big)\cdot a^2 \le |a^2 - y^2|$ powering both summable derivative majorants, plus the pointwise interval glue (`SideLobeConcavity`, `OnePeakPerLobe`) — one peak per lobe, kernel-checked; moreover Φ is continuous on the real axis (uniform products on compacts, `RvachevProductContinuity`), so each side lobe carries *exactly one* maximizer of $|\Phi|$ and the central maximizer is exactly the origin;

- **the unconditional fluctuation record**: exact centering $\int_0^1 \log|P_n| = 0$, the L^1 scale $\int_0^1 |\log|P_n|| \le (\pi/2)\sqrt{n}$ by the quadratic Cauchy–Schwarz trick (`LogProductMoments`); $\kappa_0 = 1$ *in measure* — for every $\varepsilon > 0$, $\operatorname{vol}\{t : |\log|P_n|| \ge \varepsilon n\} \le (\pi^2/4\varepsilon^2)/n \to 0$ — and the uniform-in-n tightness $\operatorname{vol}\{t : |\log|P_n|| \ge K\sqrt{n}\} \le \pi^2/(4K^2)$ at the CLT scale, both by Chebyshev against the exact variance (`KappaZeroInMeasure`, `FluctuationTightness`); and the baseline decay $\|\Phi(x)\| \le 1/(\pi|x|)$ on the whole real axis (`BaselineDecay`);
- **the zero order**: the dyadic lattice hits the integer $m \ge 1$ exactly $v_2(m)+1$ times (`LatticeFiber`), and $\|\Phi(x)\|/|x-m|^{v_2(m)+1}$ converges along the punctured neighborhood of m to an explicit strictly positive constant (`ZeroOrderTail`, `PhiZeroOrder`) — the multiplicity statement of `prop:canonical` in quantitative local form, and the local half of the valley law. Its two extremes are corollaries: Φ has *simple* zeros at odd integers and zeros of order $j+1$ at 2^j, so the order is unbounded along the dyadic integers (`ValleyZeroOrders`);
- **the sign of Φ, and it is Thue–Morse**: the split product shows Φ is real on the whole real axis and, off the zero lattice, $\Phi(x) = (-1)^{\#\{a \le \lfloor x \rfloor\}}\|\Phi(x)\|$ (`PhiRealSign`); the exceptional block depends only on $\lfloor\cdot\rfloor$, hence is constant along a lobe, and decomposes into the fibers over $1,\dots,m$, giving one sign per lobe with $\#\{a \le m\} = \sum_{k\le m}(v_2(k)+1)$ (`LobeSignCount`). Legendre's theorem in digit-sum form evaluates that count as $2m - w(m)$, so

  $$\Phi(x) = (-1)^{w(m)}\,\|\Phi(x)\|, \qquad x \in (m, m+1),$$

  with w the binary digit sum (`ThueMorseLobeSign`) — the same Thue–Morse sign that defines the signed global extension $F = \sum(-1)^{w(n)}\operatorname{up}(\cdot - 2n - 1)$ governs the lobe signs of its Fourier transform, and it is self-similar under doubling of the frequency index (`LobeSignSelfSimilarity`);
- **the renormalization identity** $\Phi(z) = \operatorname{sinc}(\pi z)\,\Phi(z/2)$ and its n-fold shell form (`RenormalizationIdentity`);
- **the global κ_∞ envelope**: there is a constant C with $\|\Phi(x)\| \le C\,E_{\kappa_\infty}(x)$ for every $x \ge 1$ (`GlobalDecayEnvelope`). The peak-ray gauge identity is proved at an *arbitrary* base point — its k-linear terms cancel precisely when $\kappa = \kappa_\infty$, so the identity characterizes the exponent — and is combined with the shell bound, which already held along every dyadic ray, and a compact-window maximum of $\|\Phi\|/E_{\kappa_\infty}$ on $[1,2]$. With the attainment along $2^k \cdot (2/3)$ this pins κ_∞ from both sides as *the* envelope exponent of the Fourier decay;
- the lacunary L^1 mean, sharpened from the Gelfond-inherited $(\sqrt{3}/2)^n$ to $\int_0^1 \prod_{j<n}|\sin(\pi 2^j t)|\,dt \le 2^{-n/2}$ by Cauchy–Schwarz against the exact second moment 2^{-n} (`LacunaryMeanSharp`).

The sentence above about the exact variance is therefore superseded: the Clausen expansion in L^2 is formalized, the one-peak profile statement is now a theorem of the development rather than a target, and of the fluctuation program only the limit theorems themselves (the martingale central limit theorem and iterated-logarithm law), the Ruelle–Perron–Frobenius eigenfunction *existence*, and the Hoischen approximation import remain outside the formalized perimeter, all blocked on machinery currently absent from Mathlib.

8 Verdict

1. **On the mathematics.** The second wave completes the problem. The one genuinely open question — unrestricted (Q2) — is settled affirmatively by two independent, correct constructions (Documents 7 and 8), and the impossibility side is sharpened to bounded-distortion gauges (Document 7). The final answer to [1] is the composite map of section 3.3.

2. **On the documents.** All four are of high quality and essentially correct; their disagreements with one another are nil and their corrections of the first audit are, in every case examined, fair. Document 6 is the cleanest exposition of the fluctuation theory; Document 7 carries the strongest new mathematics; Document 8 independently confirms it in a form best suited to formalization; Document 5 is the most careful about scope and located the profile features first.

3. **On the first audit.** Its theorems all stand; its numerics stand except one constant's seventh figure; two of its prose claims were over-broad and are corrected here and, by patch, in place. The episode is a textbook case for the repository's policy that summary sentences must not outrun the theorems they summarize.

4. **On numerical hygiene.** Three of the five errata found in the wave are last-digit prints exceeding float64 pipeline precision. The forty-digit values of section 5.1 are now the corpus reference; future documents should quote at most as many digits as their own pipeline certifies.

A Reference constants after the second wave

Best current values, their sources, and status. "Proved" means the defining object is exact and the decimal is a stable high-precision evaluation of it; only ϱ_1-derived quantities inherit the caveat that ϱ_1 itself has a rigorous enclosure ($0.66130 < \varrho_1 < 0.66135$ [3]; Sturm certificate $(0.66126807, 0.66134891)$, Documents 4/6) much wider than its computed precision.

constant	value	status
$a = \log 2$	0.69314718055994530942	exact
κ_0	3.15149612947231879805	exact ($\frac{3}{2} + \log_2 \pi$)
κ_1	2.74807001487133267392	from ϱ_1 (theorem 5.4)
κ_2	2.65149612947231879805	exact ($\log_2 2\pi$)
κ_∞	2.35901487911174070735	exact ($\frac{3}{2} + \log_2(\pi/\sqrt{3})$)
ϱ_1	0.66132260206056561507352912	collocation, N-stable to 10^{-38}
A_1	0.09126612413158821228656	spectral = direct to 16 digits
B_1	0.06505923504037692143099	spectral = direct to 16 digits
$A_1^{\log}$	0.09386063575678126354068	$= B_1/a$
C_*	0.13912977473482934529458	exact ray, 38-digit constancy
$\beta = \log(\sqrt{3}/2)$	-0.14384103622589046372	exact
$c = (a/\pi)\sqrt{3/2}$	0.27022231973336534092	exact
ϱ_2	1/2	exact
ϱ_3	0.394896632370234	collocation
ϱ_4	0.320194101601104	collocation
ϱ_8	0.161244458899776	collocation
ϱ_{16}	0.050071935712638	collocation
$H(1)$	0.55377127588881009534	30-digit product (first audit)
σ^2 per level	$\pi^2/4$	proved (three routes)
LIL constant	$\pi/\sqrt{2}$ ($\sqrt{n \log\log n}$ norm.)	proved; published π is the erratum
profile range	$\supseteq [0.91627, 1.11814]$	finite-level; extremes uncertified

B Verification scripts

- `verification_scripts/stage7_spectral.py` — 40-digit collocation: ϱ_1 ($N = 48, 64$), A_1, B_1, $A_1^{\log}$, C_* with the ray-constancy test, the q-spectrum, Document 6's convergence table, Document 8's constants appendix, $\mathrm{Var}(S_{1000})$, and $\int_0^{\pi/2} \log^2 \tan = \pi^3/8$. Writes `constants.json`.

- `verification_scripts/stage8_profile.py` — finite-level profile with integer-exact dyadic argument reduction: direct $A_1^{(n)}$, $A_1^{\log(n)}$ for $n = 13, \ldots, 20$; Document 5's feature values by exact partial-cell quadrature; Document 7's level-18/19 extremes; Document 6's sixteen-point table.
- `verification_scripts/stage8b_feature_zoom.py` — data for figure 1.

All scripts were run on 26–27 August 2026; their complete output is reproduced in the repository history alongside this file's commit.

References

[1] V. Reshetnikov, *Asymptotic decay rate of an infinite product of sinc functions*, Mathematics Stack Exchange question 2745497 (2018); repository transcription in `drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/`.

[2] *The Decay Envelope of the Rvachev Sinc Product: A Comparative Audit*, second-wave report, this repository, `drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/`.

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